

“Data Preprocessing in Machine Learning”



Data Cleaning

- handling missing values
- removing duplicates
- correcting inconsistencies



Data Transformation

- scaling
- normalization
- log transforms



Feature Engineering

- feature creation
- feature extraction
- encoding categorical variables



Dimensionality Reduction

- PCA
- t-SNE
- reducing noise and redundancy



Data Splitting

- train/validation/test separation with proper ratios

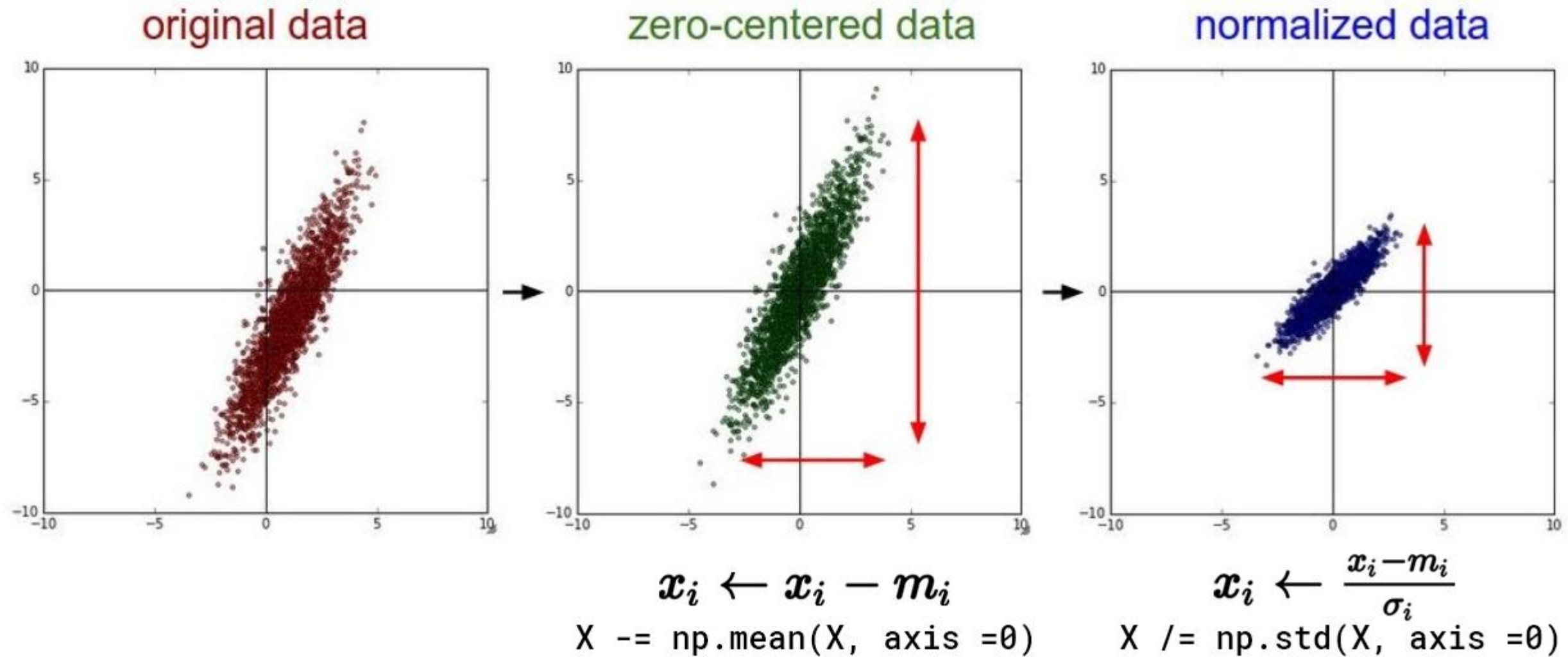
Data preprocessing

A Complete Lecture on Recurrent Neural Networks

Undergraduate Engineering — Neural Networks & Deep Learning

Normalizing The Data

- Assume $X_n \times d$ is a data matrix, with each sample represented in a row
- i is the index of dimension ($i = 1, \dots, d$)

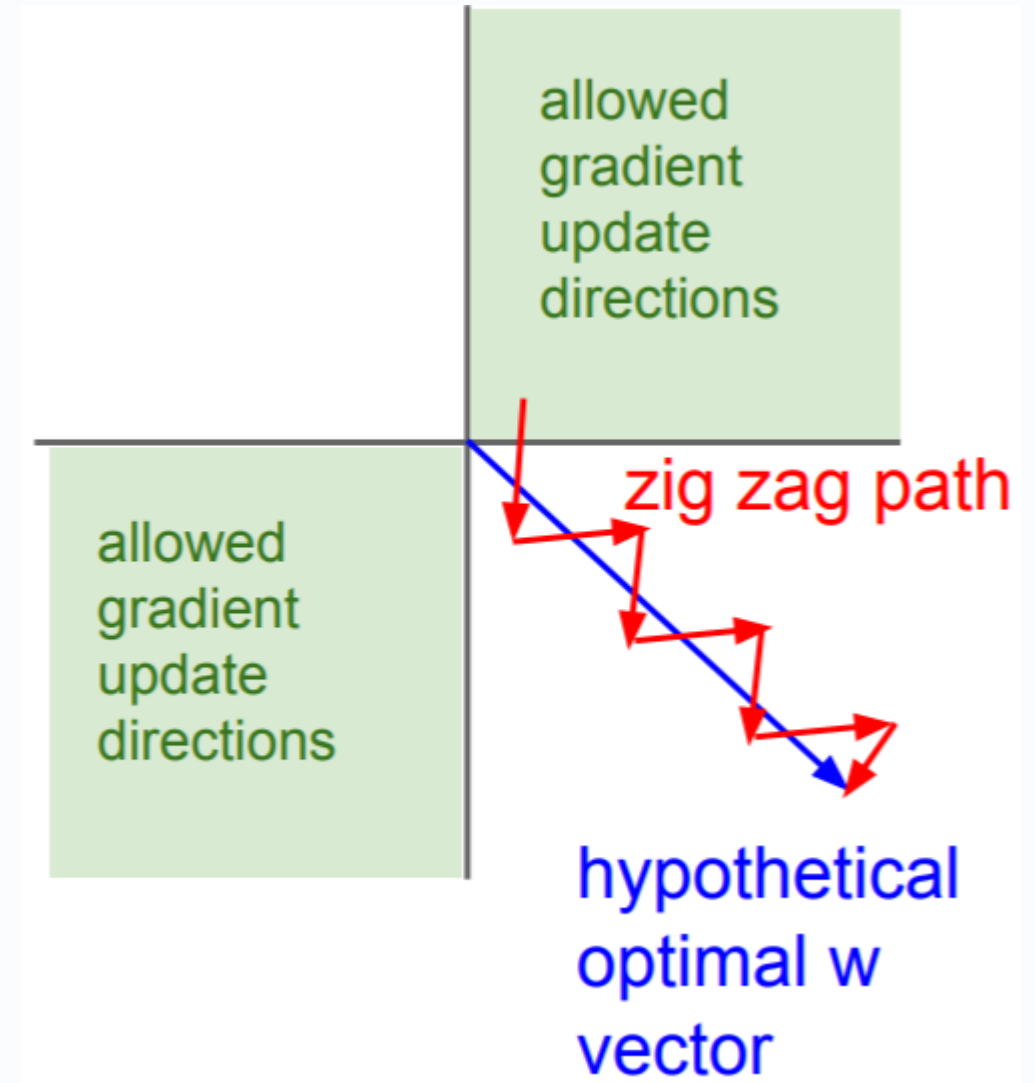


Normalizing The Data

- Example: consider CIFAR-10 example with [32,32,3] images
- 3 different approaches:
 - 1 Subtract the mean image (e.g., AlexNet)
mean image = [32,32,3] array
 - 2 Subtract per-channel mean (e.g., VGGNet)
mean along each channel = 3 numbers
 - 3 Subtract per-channel mean and divide by per-channel std (e.g., ResNet)

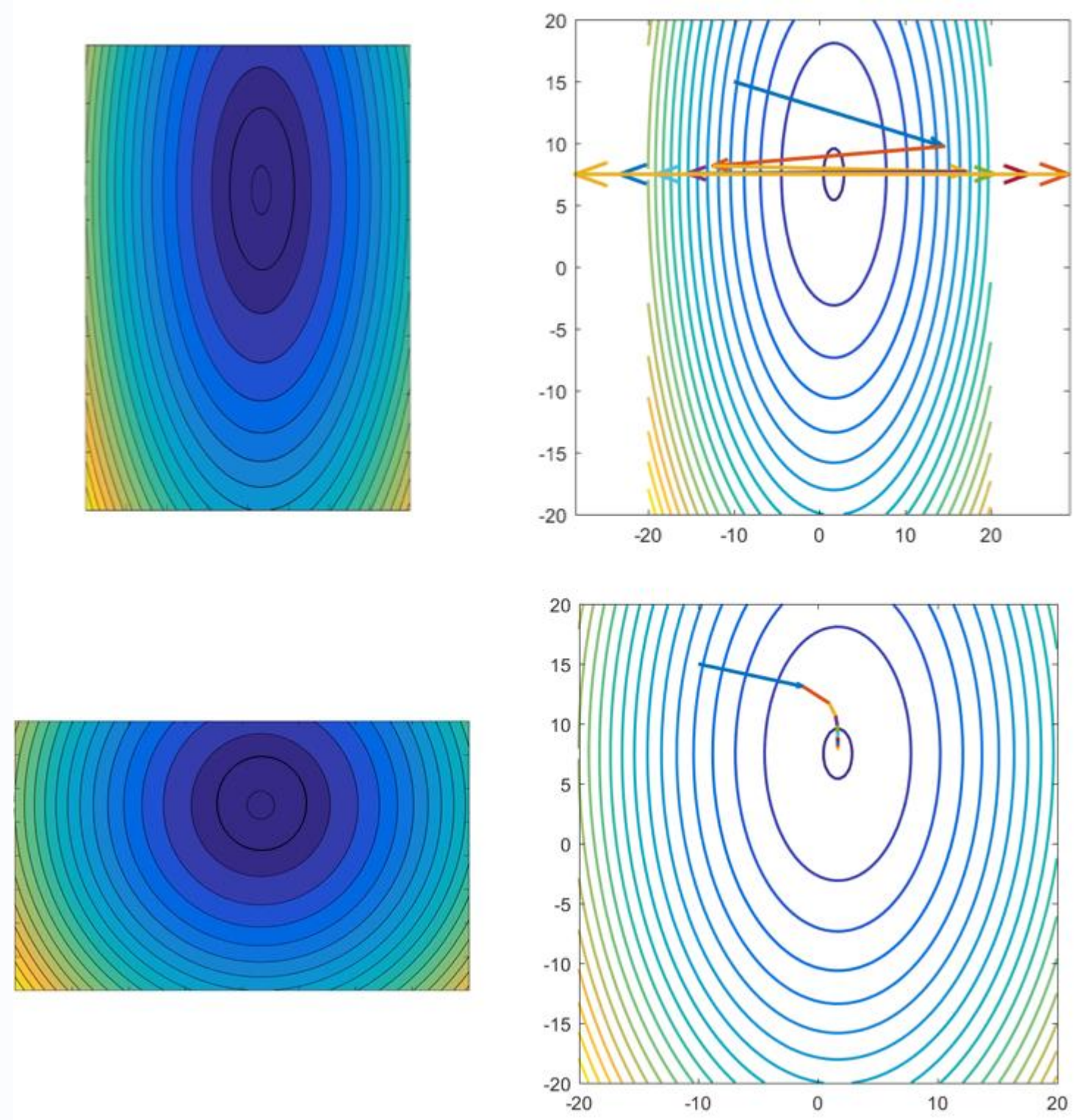
Normalizing The Data

- Reminder: sigmoid
- Consider what happens when the input to a neuron is always positive
- What can we say about the gradients on w ?
 - Always all positive or all negative
 - Leads to zig-zag dynamics in the gradient updates for the weights (in red) if the optimal w vector was the blue one
- This is also why zero-mean data is preferable.



Normalizing The Data

- Wide variation in the range of feature values
 - Leads to poor conditioning
- Results in noisy movement of gradient
- Normalized data:
 - Enhances the geometry of the loss function
 - Mitigates issues of poor conditioning
 - Accelerates convergence during training



Normalizing The Data

- While standardizing inputs is common practice, transformations like variance normalization, PCA, and whitening are used more selectively.

